

TRANSACTIONS AND SCHEDULING

Final RTL evidence | 2026-09-30 | Group 2/3: seven named benches

BENCH / OBSERVED LOG COUNTS

REFERENCE CHECK / COVERAGE BOUNDARY

sar_trial_tb

4,096 distinct fine / 512 coarse codes
stall + cancellation checks completed
Observed: SAR_TRIAL_COMPLETE

Ideal comparator against target code;
stalled trial hold, done and cancellation.
No comparator noise or metastability model.

structural_adc_tb

438 outputs; 7,680 checked clock ticks
438 latency checks; +11 cycles (P=7)
Observed: STRUCTURAL_ADC_COMPLETE

Physical-switch 256-bit scoreboard;
code/flags/ID, bank ownership and latency.
Behavioral analog boundaries; 3 modes.

structural_protocol_tb

120 frames: 57 launches / 63 drops
1,920 checked ticks; 2 decode perturbations
Observed: STRUCTURAL_PROTOCOL_COMPLETE

Expected accept/drop schedule;
captured context + edge-stable enables.
Two phase-decode perturbations are injected.

recon_ppa_latency_tb

P=5/6/7: 32/32/32 samples
latency 15/13/11; request interval 16 cycles
Observed: RECON_PPA_PROFILE_PASS stages=7 samples=32

Per-profile code/ID scoreboard;
busy-release and completion timestamps.
Clock intervals, not measured nanoseconds.

review_recon_protocol_tb

Total check count: not logged
directed completion marker present
Observed: REVIEW_RECON_PROTOCOL_COMPLETE

Directed unconfigured/cancel/error
scenarios; ID/flags and prior-output hold.
Completion only: no total check count logged.

review_top_protocol_tb

1,108 check calls; 38 reported outputs

Observed: REVIEW_TOP_PROTOCOL_COMPLETE

Top-port configuration and result oracle;
readback, rejection, epochs and IDs.
38 outputs is the TB's printed count.

review_config_tb

624 observed output-valid events
total assertion count: not logged
Observed: REVIEW_CONFIG_COMPLETE

Configuration completeness/collision
and atomic epoch guards; dither-bin checks.
624 outputs are not 624 independent tests.

Counts retain their own units. No combined 'total coverage' score. All 21 log and TB hashes verified.

Source: docs/evidence/20260930/final_rtl/ | Zero-delay RTL logs, not STA or analog performance.